

PR. Final Exam

Q.1b Data Analysis

- Data Analysis is the process of Collecting, cleaning, transforming and examining data to find useful information. It helps identify patterns, trends and relationships in a dataset and support better decision-making.

Q.2b Data Acquisition.

- Data Acquisition is the process of Collecting data from different sources for Analysis. Data may be obtained from files, database, website or APIs. In this project, CSV, JSON and SQLite data sources are used.

Q.3b Data Understanding

- Data Understanding is the process of examining a dataset before analysis. It includes checking the number of rows and columns, data type, missing values, duplicate records and basic statistical information.

Q.4b Exploratory Data Analysis (EDA)

- EDA is the process of exploring a dataset using statistics and visualization. It helps understand the distribution of variables, identify patterns, detect unusual values and peaking in consistency.

Q.5b Missing Value Imputation.

- Missing Value Imputation means replacing missing observation with suitable values instead of leaving them empty. Common approaches include mean or most frequent imputation, random sample imputation, KNN imputation and iterative imputation.

Q.76

Outliers

- Outliers are observations that are unusually high or low compared with most other observations. They can affect statistical analysis and machine learning models. Common detection methods include Z-score, IQR and percentile-based methods.

Q.81

Z-Score

- Z-score measures how far a value is from the mean in terms of standard deviation. A high absolute Z-score can indicate a possible outlier.

Formula: ~~$IQR = Q_3 - Q_1$~~ formula: $Z = (x - \text{mean}) / \text{standard deviation}$

Q.82

IQR

- IQR stands for Interquartile Range. It measures the spread of the middle 50% of the data and is useful for detecting outliers.

Formula: $IQR = Q_3 - Q_1$

Q.83

Feature Transformation

- Feature Transformation changes the representation or scale of variables so that they are more suitable for analysis or machine learning. Common transformations include encoding, binning and scaling.

Q.84

Encoding

- Encoding converts categorical values into numerical values so that machine learning algorithms can process them. Common methods used in this project include ordinal encoding, label encoding and one-hot encoding.